Exercise 4

Question 1

SELECT EmployeeID, Firstname Lastname, Department, Salary, ProjectID projectName, Budget status

From Employees

INNER JOIN Projects ;

ON A. EmployeeID = B. EmployeeID ;

EmployeeID	Firstname	Lastname	Departmen	Salary	Project ID	ProjectName	Budget	Status
John 1	John	Doe	IT	70 000	101	AI Development	100 000	completed
John 1	John	Doe	IT	70000	103	Cybersecurity	75 000	Pending
Bob 2	Bob	Smith	HR	60 000	102	Employee Training	50 000	Ongoing
Emma3	Emma	Johnson	Finance	75000	104	Financial Analysis	90 000	Ongoing
Alice 5	Alice	Wilson	Sales	65 000	105	Market Expansion	65 000	Completed
David 6	David	Clark	Finance	80 000	106	Risk management	80 000	Pending

2) SELECT EmployeeID, Firstname, Lastname, Department, Salary, ProjectID projectname, Budget, status

FROM Employee

LEFT Join - Project

On A. EmployeeID = B. EmployeeID ;

EmployeeID	FirstName	Lastname	ProjectID	ProjectName	Status
1	John	Doe	101	AI Development	Completed
1	John	Doe	103	cybersecurity Audit	Pending
2	Bob	Smith	102	Employee Training	Ongoing
3	Emma	Johnson	104	Financial Analysis	Ongoing
4	Michael	Brown	Null	Null	Null
5	Alice	Wilson	105	Market Expansion	Completed
6	David	Clark	10	Risk Management	Pending

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3. SELECT ProjectID, ProjectName, Budget, Status, EmployeeID, FirstName
LastName, Department, Salary
FROM Employees
RIGHT JOIN Project
ON A. EmployeeID = B. EmployeeID ;

ProjectID	ProjectName	Budget	~~Budget~~	Status	EmployeeID	Firstname	Lastname	Dept	Salary
101	AI Development	100 000		Completed	1	John	Doe	IT	70 000
102	Employee Training	75 000		Ongoing	2	Bob	Smith	HR	60 000
103	Cybersecurity Aud	90 000		Pending	1	John	Doe	IT	70 000
104	Financial Analysis	90 000		Ongoing	3	Emma	Johson	Finance	75 000
105	Market Expansion	65 000		Complete	5	Alic	Wilson	Sales	65 000
106	Risk management	80 000		Pend.	0	David	Clark	Finance	80 000

4

SELECT EmployeeID, FirstName, Lastname, Department, Salary, ProjectID
ProjectName, Budget, Status
FROM Employees
FULL OUTER JOIN Project
ON A.EmployeeID = B. EmployeeID ;

EmployeID	FirstName	LastName	ProjectID	ProjectName	Status
1	John	Doe	101	AI Development	Complete
1	John	Doe	103		
2	Bob	~~Deesmith~~	102	Cybersecurity Au	ongoing
3	Emma	~~Smith~~ Johnson	104	Employee Training	ongoing
4	Michael	~~Johnson~~ Brown	~~104~~ NULL	Financial Analys	
5	Alice	~~Brown~~ Clark Wilson	~~NULL~~ 105	~~NULL~~ Market Expansion	completed
6	David	Clark	106	Risk managem	pending

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Question 5

SELECT CITY AS Location
FROM Employees
UNION
SELECT status AS Location
FR Projects;

LOCATION	STATUS
New York	Completed
Los Angel	Ongoing
Toronto	Ongoing
London	
Sydney	Completed
Completed	

Question 6

SELECT CITY CITY AS Location
UNION ALL
SELECT Status AS Location
Projects;

Location	Status
New York	Completed
Toronto	Ongoing
London	Pending
Sydney	Completed

Question 7

SELECT EmployeeID, Firstname, LastName, Department Salary
From Employees
WHERE Salary > 70 000;

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Question 5 & 6

EmployeID	FirstName	LastName	Department	Salary
3	Emma	Johson	Finance	75 000
4	Michael	Brown	IT	72 000
6	David	Clark	Finance	80 00

Question 8

```
SELECT   EmployeeID, FirstName, LastName, Department, Salary
FROM     Employees
WHERE    Department IN ('IT', 'Finance');
```

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70 00
3	Emma	Johson	Finance	75 000
4	Michael	Brown	IT	72 000
6	David	Clark	Finance	80 000

Question 9

```
SELECT   ProjectID, ProjectName, Budget, Status
FROM     Project
WHERE    Status = 'Complete';
```

ProjectID	ProjectName	Budget	Status
102	EmployeeTrainin	50 000	Ongoing
103	Cybersecurity	75000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk management	80 000	Pending

Question 10

```
SELECT   ProjectID, ProjectName, Budget, status
FROM     Projects
WHERE    Budget > 70 000 AND Status != 'Completed';
```

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ProjectID	ProjectName	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk management	80 000	Pending

Question 11

```
SELECT  ~~Project~~ EmployeeID, FirstName, LastName, Department, Salary, City
FROM  Employees
WHERE  City IN ('New York', 'Toronto')
ORDER BY Sallary DESC;
```

EmployeeID	FirstName	LastName	Department	Salary	City
6	David	Clark	Finance	80 000	New York
3	Emma	Johnson	Finance	75 000	Toronto
1	John	Doe	IT	70 00	New York

Question 12

```
SELECT  Top 3 EmployeeID, FirstName, LastName, Department, Salary
FROM  Employees
ORDER BY Salary DESC;
```

EmployeeID	FirstName	LastName	Department	Salary
6	David	Clark	Finance	80000
3	Emma	Johnson	Finance	75000
4	Michael	Brown	IT	72000

Question 13

```
SELECT  Department, Sum (Salary) AS TotalSalary
FROM  Employees
GROUP BY Department
ORDER BY TotalSalary DESC;
```

Department	Total Salary
Finance	155 000
IT	142 000
Sales	65 000
HR	60 000

Question 14

```
SELECT City, Avg (salary) As AverageSalary
FROM Employees
GROUP BY City
HAVING Avg (salary) > 65000;
```

City	Average Salary
New York	75 000
Toronto	75 000
London	72 000

Question 15

```
SELECT Department, Count (EmployeeID) As EmployeeCount
FROM Employees
GROUP BY Department
HAVING COUNT (EmployeeID) > 1;
```

Department	Employee Count
IT	2
Finance	2

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~~Question 16~~

~~SELECT Department, Count (EmployeeID) As EmployeeCount~~

~~FROM Employees~~

~~GROUP BY Department~~

~~HAVING COUNT (EmployeeID)~~

Question 16

```
SELECT Status, COUNT (ProjectID) As projectCOUNT
FROM Projects
GROUP BY Status
HAVING COUNT (ProjectID) >= 2;
```

Status	Project
Completed	2
Ongoing	2
Pending	2

Question 17

```
SELECT Employee, FirstName, LastName, Sum (Budget) As TotalProjectB
FROM Employee
JOIN Project ON A.EmployeeID = B.EmployeeID
GROUP BY EmployeeID, FirstName, LastName
HAVING Sum (Budget) > 150000
```

EmployeeID	FirstName	LastName	TotalProjectBudg
1	John	Doe	175 000

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